

UT MISSISSAUGA, ON, Canada

WMO: 716245

Lat: 43.5525N

Lon: 79.6665W

Elev: 132

StdP: 99.75

Time Zone: -5.00 (NAE)

Period: 00-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.4	-15.5	-21.4	0.6	-17.5	-18.4	0.8	-14.5	13.0	-2.3	11.6	-3.1	3.6	250	0.562

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.9	32.0	24.4	30.1	23.3	28.5	22.3	25.9	29.8	24.7	28.5	23.5	27.1	4.7	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.9	20.3	27.7	23.6	18.7	26.9	22.4	17.3	25.5	81.0	30.0	75.7	28.5	71.0	27.2	32.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.9	10.2	8.9	-21.4	34.8	3.5	1.7	-23.9	36.0	-26.0	37.0	-27.9	37.9	-30.5	39.2		
			-21.7	27.3	3.4	1.1	-24.1	28.1	-26.1	28.7	-28.0	29.4	-30.5	30.2		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.7	-4.4	-4.0	0.8	7.1	13.7	18.7	21.3	20.8	16.8	10.0	4.4	-1.6
	DBStd	10.25	5.95	5.42	5.39	4.68	4.47	3.64	2.99	2.94	3.82	4.35	4.43	4.97
	HDD10.0	1839	445	391	289	112	13	0	0	0	1	54	173	360
	HDD18.3	3792	703	624	544	339	158	39	6	10	72	260	417	618
	CDD10.0	1368	0	0	3	24	127	261	351	336	206	54	6	1
	CDD18.3	279	0	0	0	1	13	50	99	87	27	2	0	0
	CDH23.3	2543	0	0	1	23	177	504	897	699	225	16	0	0
	CDH26.7	810	0	0	0	6	53	168	306	208	67	2	0	0
Wind	WSAvg	3.9	4.6	4.9	5.0	5.0	3.9	3.0	2.8	2.8	2.7	3.4	4.0	4.7
Precipitation	PrecAvg	804	58	54	54	74	71	79	79	74	65	69	66	62
	PrecMax	1025	129	109	110	130	130	170	177	139	164	133	127	114
	PrecMin	597	24	28	21	27	10	30	20	17	18	19	21	25
	PrecStd	117	26	23	26	28	28	40	39	31	31	32	28	24
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.7	13.1	19.4	26.3	30.9	33.1	34.2	33.9	32.1	25.6	18.4	12.6
		MCWB	10.4	10.8	13.7	16.5	22.7	24.4	26.0	24.9	23.9	19.6	14.0	10.9
	2%	DB	7.9	7.8	14.8	21.8	28.0	31.0	31.9	30.8	28.5	21.8	15.4	9.3
		MCWB	6.8	5.9	10.5	13.8	20.9	23.4	24.7	24.0	22.3	17.8	12.8	8.4
	5%	DB	5.1	5.0	11.5	18.5	24.8	28.5	30.0	28.9	25.8	18.9	13.7	6.9
		MCWB	4.1	3.6	8.1	12.1	18.4	22.3	23.3	22.9	20.6	16.1	11.6	6.1
	10%	DB	3.4	3.2	8.3	15.4	22.2	26.3	28.1	27.1	23.5	16.6	11.3	5.0
		MCWB	2.6	2.1	5.8	10.1	16.5	20.5	22.2	21.7	19.6	14.8	9.6	4.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.1	11.3	15.1	17.9	24.7	26.6	28.2	27.4	25.9	21.1	15.3	11.4
		MCDB	11.7	12.2	18.1	22.6	30.3	30.3	31.9	30.0	29.8	23.9	16.7	12.3
	2%	WB	6.9	6.1	11.4	15.5	21.9	24.7	25.9	25.5	23.4	18.7	13.6	8.5
		MCDB	7.9	7.3	13.5	19.9	26.4	29.5	29.8	28.7	26.5	20.7	15.0	9.2
	5%	WB	4.3	3.8	8.4	12.8	19.6	23.2	24.6	24.2	21.8	16.7	11.6	6.2
		MCDB	4.9	4.7	11.0	17.4	23.1	27.4	28.4	27.4	24.4	18.2	13.5	6.9
	10%	WB	2.6	2.1	5.9	10.4	17.6	21.5	23.3	23.0	20.5	15.2	9.7	4.2
		MCDB	3.3	3.0	8.0	14.7	21.2	24.7	26.7	26.2	22.7	16.6	11.0	4.9
Mean Daily Temperature Range		MDBR	8.1	9.1	9.9	11.6	12.3	11.8	11.9	11.6	11.4	9.7	8.9	7.0
	5% DB	MCDBR	9.0	10.8	14.4	17.7	16.0	15.0	14.0	13.5	14.7	12.9	12.2	8.8
		MCWBR	7.9	8.8	9.8	10.6	9.2	8.7	7.5	7.4	8.3	8.6	9.2	7.6
		MCDBR	8.6	10.3	13.2	15.9	14.4	13.8	12.2	11.5	12.3	11.6	11.1	8.7
		MCWBR	7.9	8.9	9.8	10.9	9.4	8.8	7.8	7.4	8.0	8.6	9.0	7.9
Clear-Sky Solar Irradiance	taub	0.295	0.325	0.334	0.380	0.406	0.438	0.430	0.416	0.384	0.363	0.316	0.299	
	taud	2.404	2.301	2.395	2.303	2.281	2.217	2.265	2.293	2.362	2.402	2.494	2.442	
	Ebn at Noon	836	862	906	884	867	837	840	839	840	811	808	796	
	Edh at Noon	76	100	104	124	130	139	132	124	107	91	70	68	
All-Sky Solar Radiation	RadAvg	1.42	2.16	3.44	4.42	5.36	5.85	5.86	5.14	4.13	2.48	1.56	1.13	
	RadStd	0.14	0.20	0.35	0.48	0.56	0.42	0.43	0.26	0.39	0.22	0.20	0.12	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (50 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page